MIL-F-16377/63(SH)
11 April 1983

SUPERSEDING
MIL-F-24361A(SHIPS)
6 February 1970

MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, FLOODLIGHTS, PORTABLE, 600 WATTS, 120/28 VOLTS, DRIPPROOF 15 DEGREES, SYMBOL 311

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

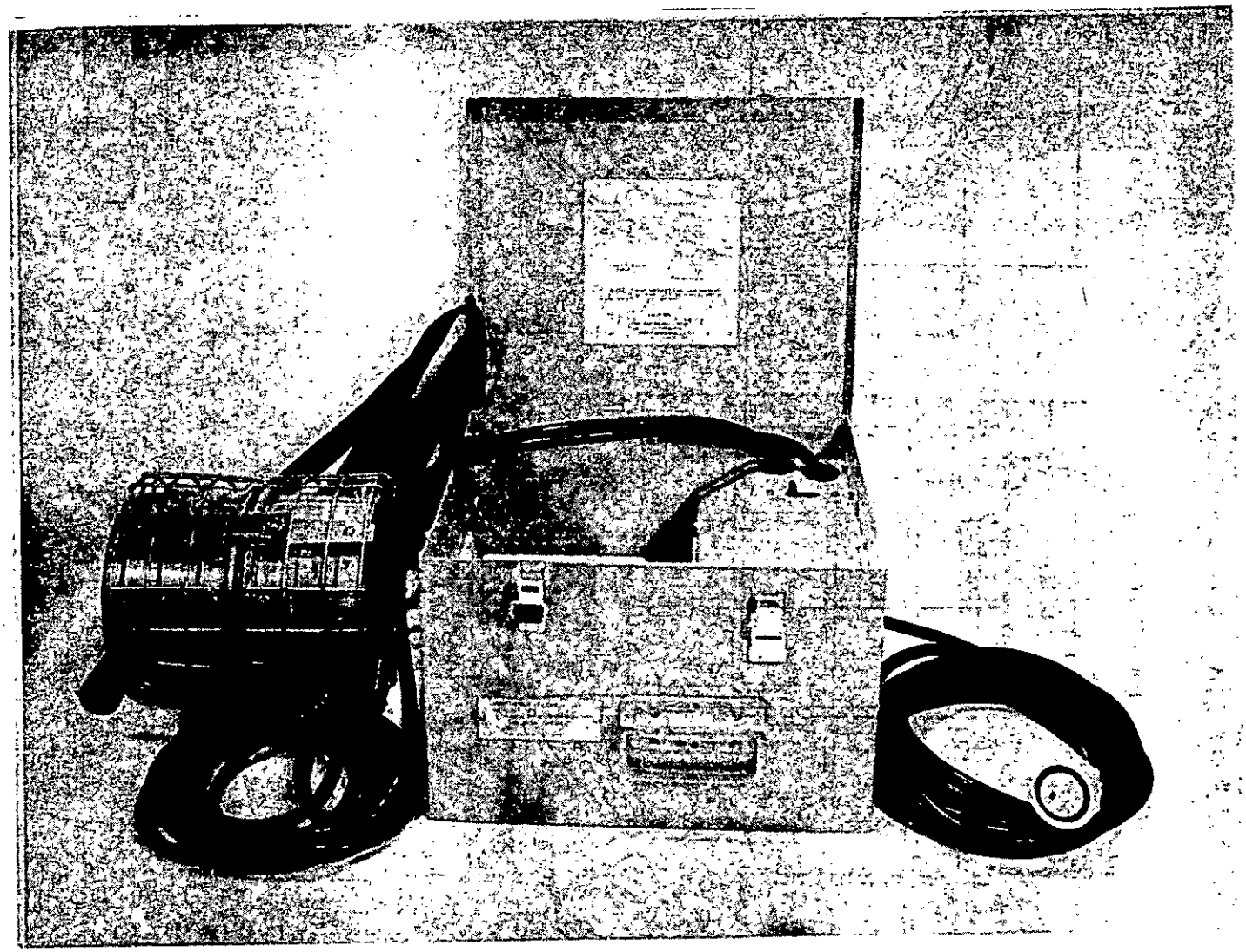


FIGURE 1. Typical view of symbol 311.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

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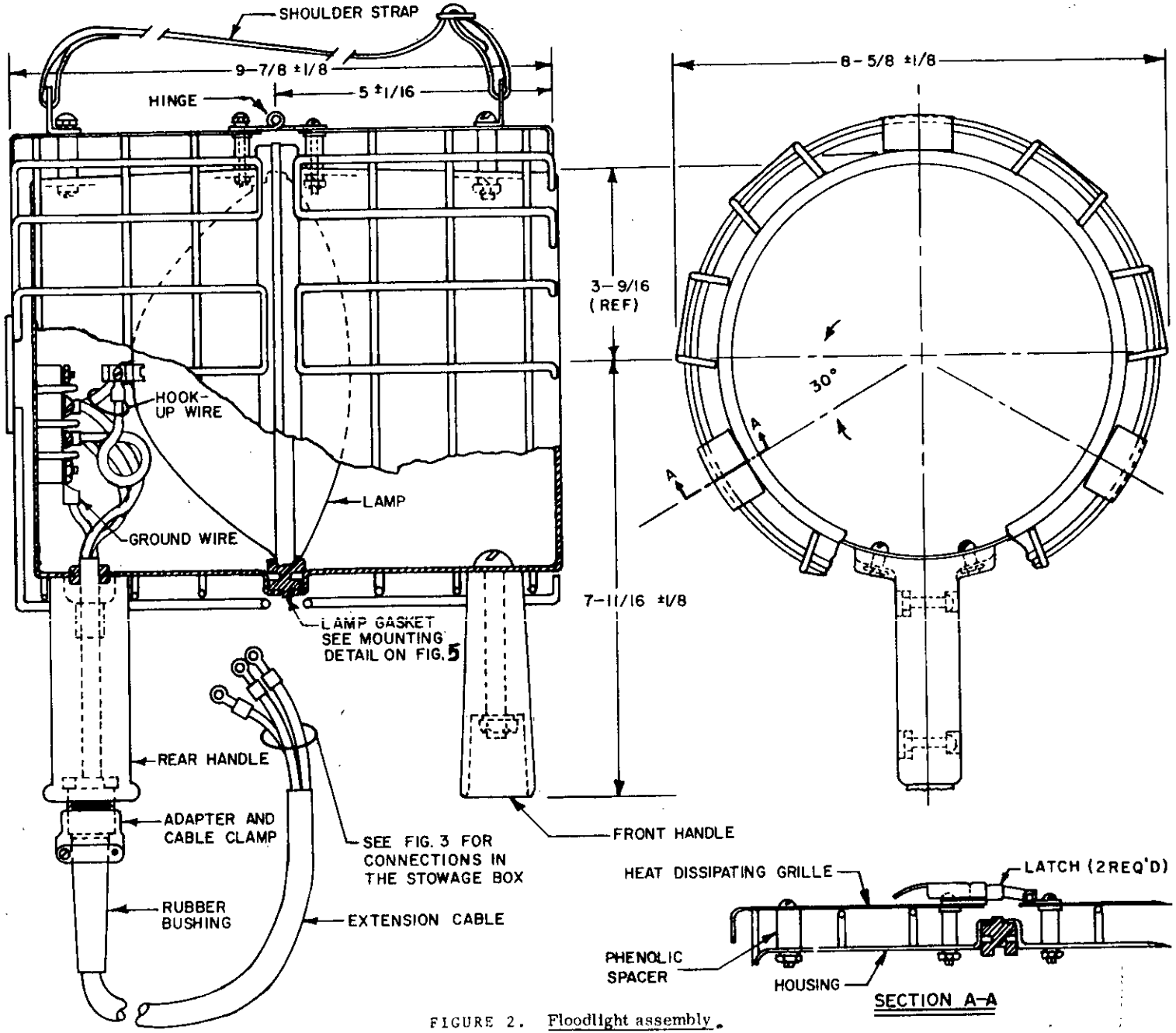


FIGURE 2. Floodlight assembly.

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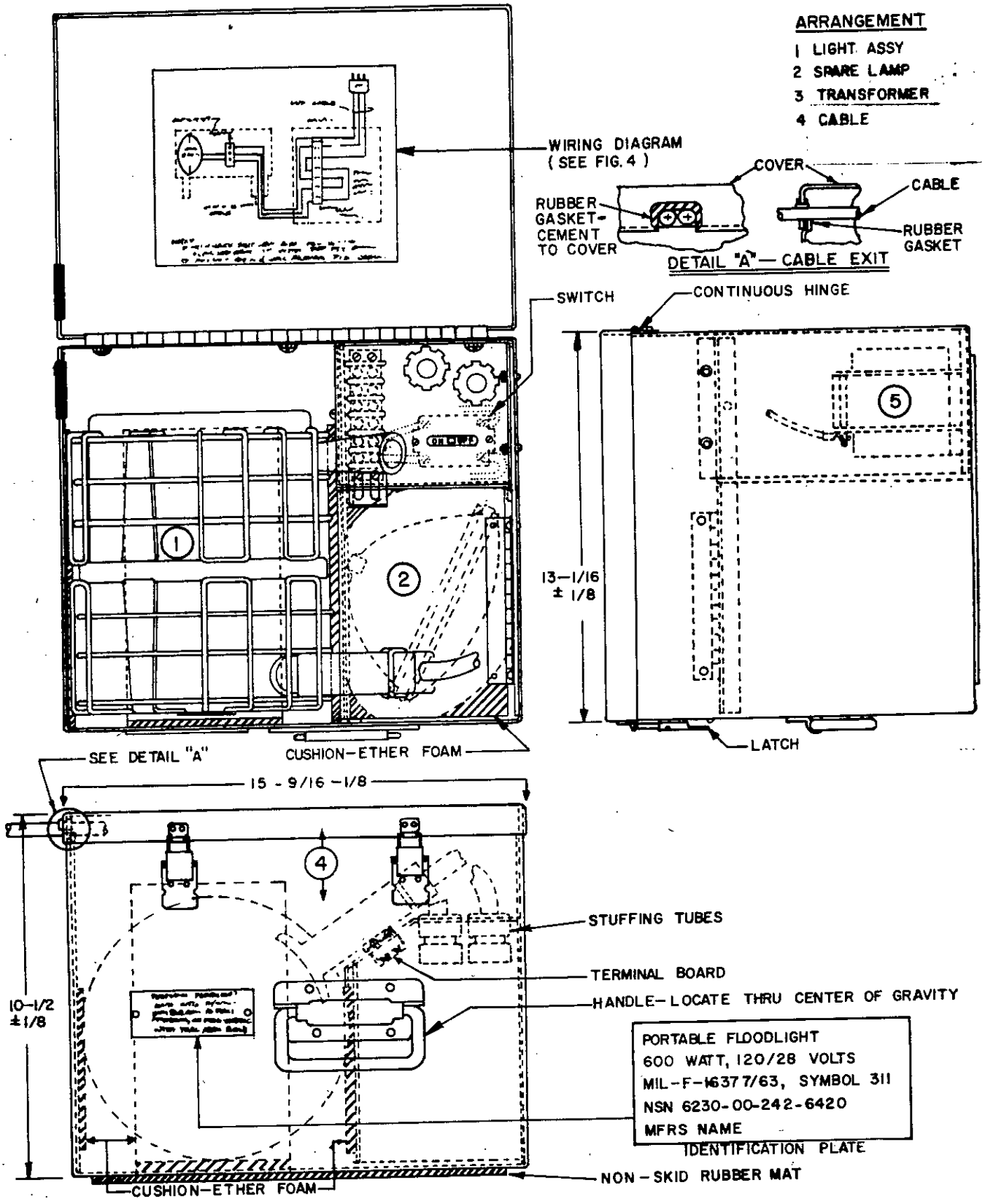
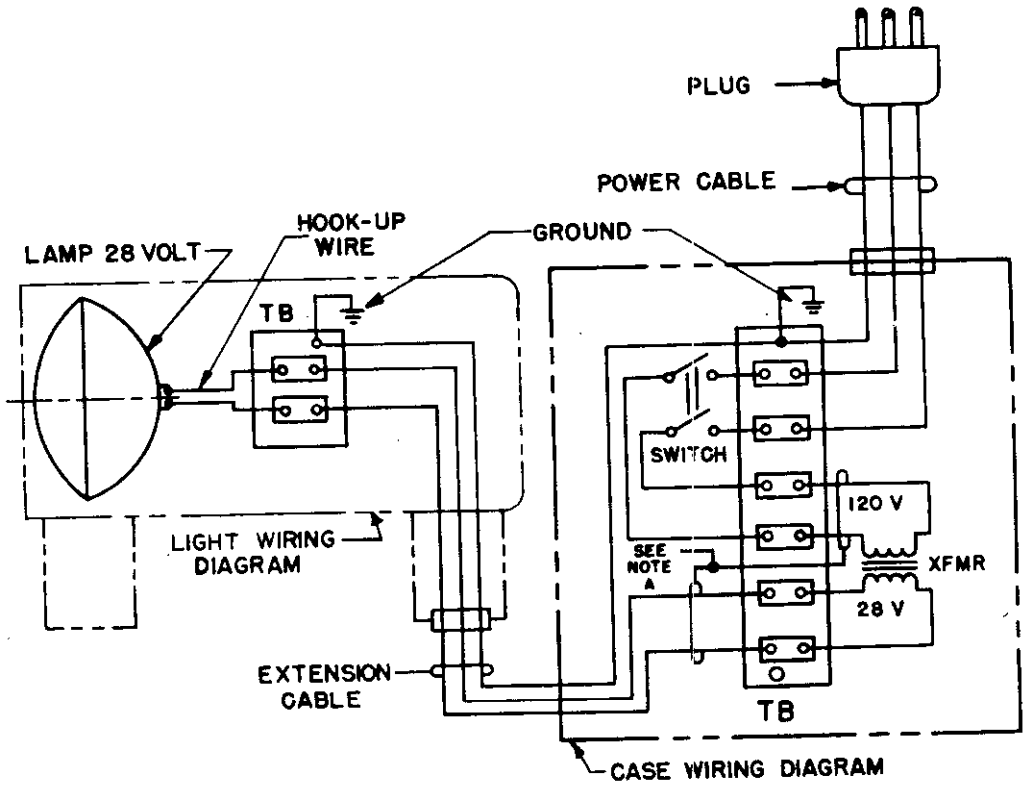


FIGURE 3. Stowage box assembly.

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NOTES:

A. DISCONNECT TRANSFORMER AND CONNECT LAMP DIRECTLY TO SWITCH TERMINALS WHEN LIGHT IS USED IN 28 VOLT SYSTEM.

B. USE LAMP NSN 6240-00-242-6423 ONLY.

CAUTION
LAMP LENS MAY BE CRACKED BY SPILLAGE OF WATER OVER IT WHEN HOT

SH 9888

FIGURE 4. Wiring diagram.

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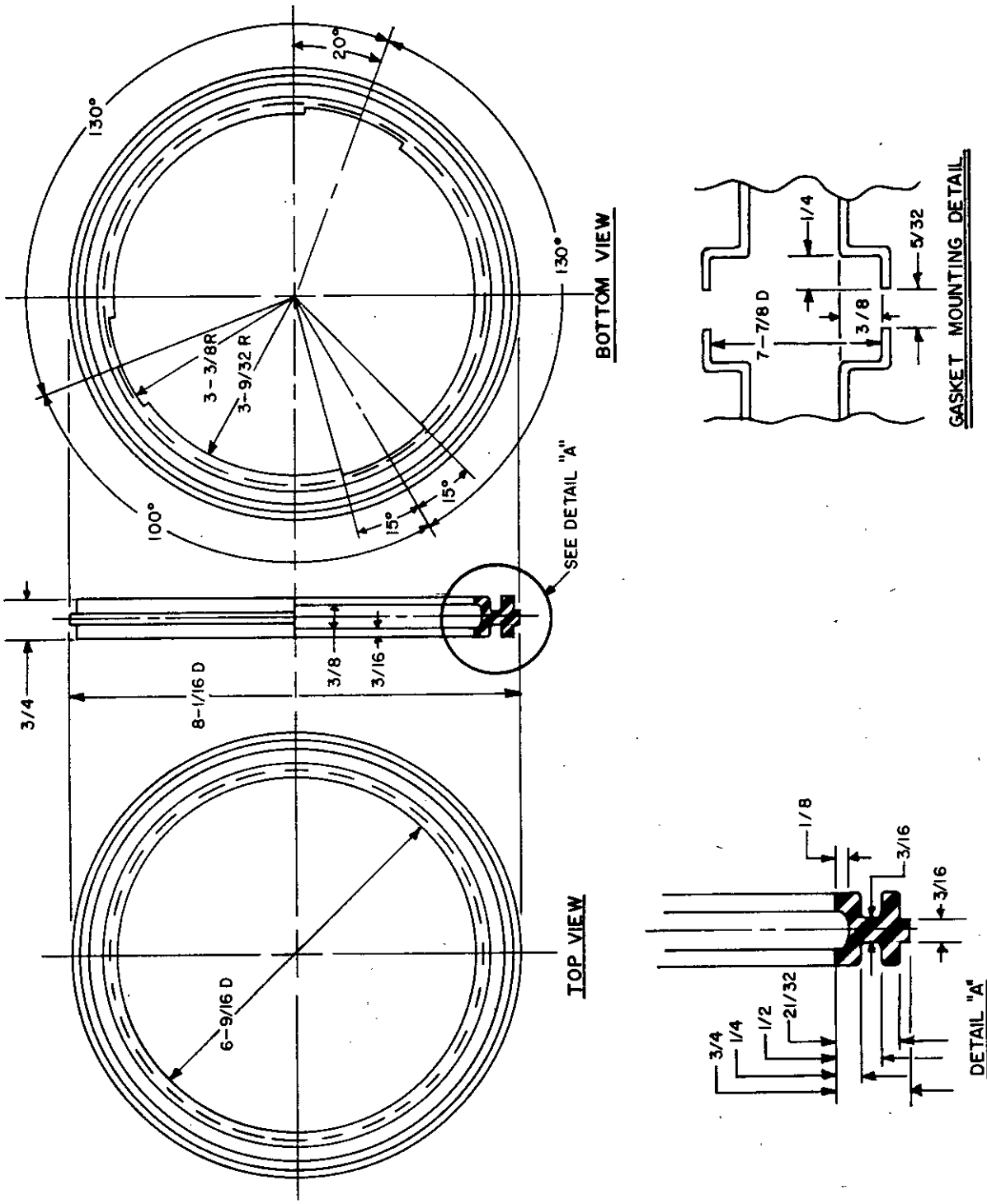


FIGURE 5. Lamp gasket and gasket mounting detail.

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REQUIREMENTS:

CLASSIFICATION:
Type: II
Class: 1

SYMBOL NUMBER:
311: Floodlight, portable, PAR56 sealed beam lamp, white.

DESIGN:
Electrical rating: 600 watts, 120/28 volts, 60 Hz.
Dimensions and configuration: See figures 1 through 5.

Floodlight assembly. The floodlight assembly shall be constructed in general conformance with figure 2. The floodlight assembly shall be so designed as to provide simple and sturdy assembly for the following parts:

(a) Housing. The housing shall be constructed of aluminum in two sections. The two sections shall be hinged and two latches shall provide for easy access to the interior of the fixture.

(b) Heat dissipating grille. A heat dissipating grille shall be provided around the outside surface of the housing to dissipate as much of the generated heat as practicable. Any portion of the floodlight which is likely to come in contact with the body shall not exceed 135° Fahrenheit (F.) when tested as specified in table I.

(c) Front handle. The front handle shall be constructed of molded thermosetting plastic of a type adapted for the purpose. The handle shall be smooth with no rough edges.

(d) Rear handle. The rear handle shall be constructed of molded thermosetting plastic of a type adapted for the purpose. The handle shall be smooth with no rough edges, and shall be designed to accommodate the cable clamp, rubber bushing, and extension cable.

(e) Cable clamp. An adapter may be provided on the bottom of the handle to mate with a cable clamp.

(f) Rubber bushing. A rubber bushing shall be installed to provide strain relief for the cable and to prevent entrance of water.

(g) Extension cable. The extension cable shall be three conductor, type THOF-4 in accordance with MIL-C-915/6. The cable shall be 15 feet plus 3 minus 0 inches long measured on the exposed cable between the cable clamp and the stuffing tube in stowage box. The third conductor (red) shall be used for grounding purposes.

(h) Lamp. Two lamps shall be furnished. One shall be installed in the floodlight and the other in the stowage box. The lamp shall be sealed beam, PAR-56, quartz-iodize, screw terminal, 600-watt, 28-volt, Industry Number Q600 PAR56/1, NSN 6240-00-242-6423, as follows:

(1)	Initial beam candlepower (approx.)	105,000
(2)	Filament construction	CC-6
(3)	Horizontal beam spread in degrees	40
(4)	Vertical beam spread in degrees	15
(5)	Hours life (minimum)	100
(6)	Finish	Fluted lens

(i) Lamp gasket. The lamp shall be secured to the housing by means of a rubber gasket in accordance with figure 5 to provide for maximum protection for the lamp. The material for the gasket shall be high temperature silicone rubber.

(j) Shoulder strap. An adjustable shoulder strap fabricated of 1 inch wide nylon webbing shall be provided to allow maximum ease of handling of the floodlight during extended periods of operation. The shoulder strap shall be adjustable to a length of 52 inches.

(k) Terminal board. A screw type terminal board shall be provided, wired as shown on figure 4.

(l) Hook-up wire. All wire shall be of stranded soft annealed copper of a cross section to provide ample and safe current-carrying capacity and mechanical strength. Only high temperature wire (200° Centigrade(C.) minimum) (with silicone outer covering) shall be used with the floodlight. All necessary internal wiring shall be furnished and installed. Connections at screw terminals shall be made with pressure-grip (solderless) connectors.

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Stowage Box. The stowage box shall be constructed as shown on figure 3. The overall size shall be kept to a minimum. The dimensions shall not exceed those shown on figure 3. The material shall be aluminum, protected against corrosion, and of sufficient gage to provide firm support for parts and compartmentation without excessive weight. A commercially available box and cover of drawn or fabricated construction adapted for the purpose will be acceptable.

(a) Compartmentation. Compartmentation shall be provided for the items specified in figure 3. Aluminum baffles and cushions of foamed plastic, rubber, or felt shall be provided to prevent movement and damage of parts in transit and stowage aboard ship. The repair parts compartment shall have a hinged cover. The transformer compartment shall have a screw-down cover.

(b) Transformer. The transformer shall be 120/28 volts. The transformer shall be firmly mounted to the box and wired as shown on the wiring diagram, figure 4. The transformer shall be capable of delivering 28 ± 1 volts to the terminals of a 600-watt lighted lamp from a 120-volt, 60 Hz, supply. The temperature rise at continuous full load shall not exceed 105°C. The transformer shall have a Class A insulation in accordance with MIL-T-16315.

(c) Power cable. A cable assembly for supplying power to the stowage box from the ship's service shall be furnished. The cable shall be three-conductor (red conductor shall be used for grounding) type THOF-4 in accordance with MIL-C-915/6. The cable shall be 25 feet, plus 3 minus 0 inches long measured on the exposed cable between the stuffing tube and plug. The cable shall be fitted at one end with a two-pole, grounded male plug, Symbol Number 1218.3 in accordance with MIL-R-2726/38.

(d) Switch. A switch shall be provided and shall be mounted and wired as shown on figures 3 and 4. The switch shall be double pole, single throw (DPST), Underwriter's Laboratories approved, and shall have a minimum rating of 25 amperes at 28 volts.

(e) Terminal board. A screw type terminal board shall be provided, wired as shown on figure 4.

(f) Stuffing tubes. Two nylon stuffing tubes in accordance with MIL-S-19622/1, size 2, shall be provided for cable entrance of the extension and power cables. The stuffing tubes shall be mounted as shown on figure 3.

(g) Handle. The stowage box shall be provided with a spring-loaded handle, mounted as shown on figure 3.

(h) Latches. At least two latches shall be provided for securing the hinged cover. Neither latch shall open under the shock and vibration.

(i) Non-skid rubber mat. A nonskid rubber mat shall be provided on the bottom of the box.

Weight. Parts shall be as small and as light as possible, but shall be sufficiently rugged to withstand rough handling so that mechanical damage or distortion will not be of such magnitude as to impair continued operation. The total weight of the floodlight shall not exceed 6 pounds. The total weight of the floodlight kit shall not exceed 42 pounds.

Designation and marking: Identification plate and label shall be furnished in accordance with MIL-F-16377. One identification plate shall be furnished as shown on figure 3. One label, type III, shall be applied on the inside surface of the cover of the stowage box and shall depict the wiring diagram and the information shown on figure 4.

Finish: Gray enamel.

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QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:
First article inspection:
Complete floodlights: See table I.
Quality conformance:
Comparison inspection: Same as first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation for delivery: See MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

GENERAL INFORMATION:
(a) Assignment of Military part numbers and general information is shown in table II.
(b) Supersession data. This floodlight test was previously covered by MIL-F-24361(SHIPS).
(c) Intended use. The floodlight kit is designed for use in search and rescue operations such as finding a man overboard, mooring to a buoy, disentangling downed flyers from their parachutes or harnesses, junk searches, or antiswimmer lighting.

TABLE I. First article and comparison inspection for complete floodlights.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	DP-15 degrees
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14	DP-15 degrees
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Grounding circuit	3.6.13.3	4.8.11	
Salt spray	3.5.15	4.8.10	
Heat dissipation	-	-	2/

1/ The stowage box, complete with parts (see 3.5.12), shall be securely strapped to a test plate and subjected to the test specified.
2/ The floodlight shall be operated at a normal voltage for 2 hours at an ambient temperature of 75° ± 5°F. At the end of the 2-hour period, the temperature at different points of the floodlight shall be recorded.

TABLE II. Assignment of part numbers and general information.

Military Part Number M16377/63-	Item name	National Stock Number
311	Complete floodlight kit	6230-00-242-6420
001	Lamp	6240-00-242-6423
002	Lampgaskets	

Preparing Activity:
Navy - SH
(Project 6230-N286)

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